

The graded 2-core vanishing law: $\text{ord}_{q=-1} G_\lambda(q) = \lfloor |2\text{-core}(\lambda)|/2 \rfloor$

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Abstract

For a partition $\lambda \vdash n$ let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$, where $s(T) = \sum_{i \in \text{Des}(T)} w_i$ with parity-twisted branch weights $w_i = 2i - 1$ if $n - i$ is odd and $w_i = 0$ otherwise. This is the graded generating function of the branch-exponent (Theorem B) descent statistic; it is generically irreducible over $\mathbb{Q}$ and has *no* product formula. I prove that its order of vanishing at $q = -1$ is governed entirely by the 2-core of λ :

$$\text{ord}_{q=-1} G_\lambda(q) = c(\lambda) := \left\lfloor \frac{1}{2} |2\text{-core}(\lambda)| \right\rfloor,$$

and I give the first nonzero Taylor coefficient at $q = -1$ in closed form via the 2-quotient. The proof rests on a single closed identity expressing $G_\lambda(-q)$ as a Schur-inner product,

$$G_\lambda(-q) = \left\langle s_\lambda, h_1^e \prod_{k: n-k \text{ odd}} (h_2 - e_2 q^{2k-1}) \right\rangle, \quad e = n \bmod 2,$$

after which the law follows from one structural fact of character theory: removing dominoes preserves the 2-core, so $\chi^\lambda(2^a 1^b) = 0$ whenever $b < |2\text{-core}(\lambda)|$. Specialising at $q = 1$ recovers the previously established fiber $G_\lambda(-1) = \chi^\lambda(w_0)$. All steps are verified computationally for every partition of $n \leq 12$.

1 Statement

Throughout $\lambda \vdash n$ and $\text{SYT}(\lambda)$ is the set of standard Young tableaux of shape λ . For $T \in \text{SYT}(\lambda)$,

$$\text{Des}(T) = \{ i \in [n-1] : i+1 \text{ is in a strictly lower row of } T \text{ than } i \}, \quad \text{comaj}(T) = \sum_{i \in \text{Des}(T)} (n-i).$$

Fix the *active set* and the branch weights

$$A = \{ k \in [n-1] : n-k \text{ odd} \}, \quad w_k = (2k-1) [k \in A], \quad s(T) = \sum_{i \in \text{Des}(T)} w_i,$$

and put $m = |A| = \lfloor n/2 \rfloor$ and $e = n \bmod 2$ (so A is the set of even positions when n is odd, the set of odd positions when n is even). The object of study is

$$G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)} \in \mathbb{Z}_{\geq 0}[q].$$

Recall that every partition has a 2-core (the unique partition with no removable domino reachable by deleting dominoes) of size $c_0 := |2\text{-core}(\lambda)|$ and a 2-quotient $(\lambda^{(0)}, \lambda^{(1)})$; one has $c_0 \equiv n \pmod{2}$ and $|\lambda^{(0)}| + |\lambda^{(1)}| = w := (n - c_0)/2$ (the number of dominoes). Set $c(\lambda) = \lfloor c_0/2 \rfloor$, so that $c_0 = 2c + e$.

Theorem 1. For every $\lambda \vdash n$,

$$\text{ord}_{q=-1} G_\lambda(q) = c(\lambda) = \left\lfloor \frac{1}{2} |2\text{-core}(\lambda)| \right\rfloor.$$

Writing $G_\lambda(q) = \sum_{k \geq 0} a_k (q+1)^k$, the first nonzero coefficient is

$$a_c = (-1)^c e_c(\{2k-1 : k \in A\}) \cdot \left(-\frac{1}{2}\right)^c \chi^\lambda(2^{m-c} 1^{c_0}),$$

where e_c is the elementary symmetric polynomial of degree c . Equivalently, via the 2-quotient,

$$a_c = (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} S(n, c), \quad S(n, c) = \pm e_c(\{2k-1 : k \in A\}) 2^{-c} f^{2\text{-core}(\lambda)},$$

a scalar depending only on (n, c) ; in particular $S(n, 0) = 1$ and $S(n, 1) = -\binom{n}{2}$.

Here $f^\mu = \# \text{SYT}(\mu)$ and $n(\lambda) = \sum_i (i-1)\lambda_i$.

2 The master identity

Let Λ be the ring of symmetric functions, s_λ the Schur functions, h_r, e_r, p_r the complete, elementary and power-sum bases, and $\langle \cdot, \cdot \rangle$ the Hall inner product, for which $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ (the Kostka number) and $\langle s_\lambda, p_\mu \rangle = \chi^\lambda(\mu)$. For a subset $R \subseteq [n-1]$ with $R = \{r_1 < \dots < r_t\}$ let $\alpha(R) = (r_1, r_2 - r_1, \dots, n - r_t)$ be the composition of n with descent set R . We use the classical fact

$$K_{\lambda, \alpha(R)} = \langle s_\lambda, h_{\alpha(R)} \rangle = \#\{T \in \text{SYT}(\lambda) : \text{Des}(T) \subseteq R\} \quad (1)$$

(Stanley, *EC2*, §7.19; the right equality is standardisation of semistandard tableaux of content $\alpha(R)$). Write $\beta_\lambda(S) = \#\{T : \text{Des}(T) = S\}$, so that $K_{\lambda, \alpha(R)} = \sum_{S \subseteq R} \beta_\lambda(S)$.

Lemma 1 (Descent enumerator). For indeterminates $\eta = (\eta_1, \dots, \eta_{n-1})$,

$$\Psi_\lambda(\eta) := \sum_{T \in \text{SYT}(\lambda)} \prod_{k \in \text{Des}(T)} \eta_k = \sum_{R \subseteq [n-1]} K_{\lambda, \alpha(R)} \prod_{k \in R} \eta_k \prod_{k \notin R} (1 - \eta_k).$$

Proof. Substitute $K_{\lambda, \alpha(R)} = \sum_{S \subseteq R} \beta_\lambda(S)$ into the right-hand side and exchange the order of summation:

$$\sum_R \left(\sum_{S \subseteq R} \beta_\lambda(S) \right) \prod_{k \in R} \eta_k \prod_{k \notin R} (1 - \eta_k) = \sum_S \beta_\lambda(S) \sum_{R \supseteq S} \prod_{k \in R} \eta_k \prod_{k \notin R} (1 - \eta_k).$$

For fixed S , writing $R = S \sqcup S'$ with $S' \subseteq [n-1] \setminus S$,

$$\sum_{R \supseteq S} \prod_{k \in R} \eta_k \prod_{k \notin R} (1 - \eta_k) = \prod_{k \in S} \eta_k \prod_{k \in [n-1] \setminus S} (\eta_k + (1 - \eta_k)) = \prod_{k \in S} \eta_k.$$

Hence the right-hand side equals $\sum_S \beta_\lambda(S) \prod_{k \in S} \eta_k = \Psi_\lambda(\eta)$. □

Proposition 1 (Master identity). As polynomials in q ,

$$G_\lambda(-q) = \left\langle s_\lambda, h_1^e \prod_{k \in A} (h_2 - e_2 q^{2k-1}) \right\rangle.$$

Proof. Set $\eta_k = (-1)^{n-k} q^{w_k}$ in Lemma ?? . Since $(-1)^{n-k} = (-1)^{w_k}$ for every k (indeed w_k is odd exactly when $n - k$ is odd), $\prod_{k \in \text{Des}(T)} \eta_k = (-1)^{\text{comaj}(T)} q^{s(T)} = (-1)^{s(T)} q^{s(T)} = (-q)^{s(T)}$, so the left side of Lemma ?? is $\Psi_\lambda(\eta) = G_\lambda(-q)$.

For $k \notin A$ we have $n - k$ even and $w_k = 0$, hence $\eta_k = 1$ and the factor $1 - \eta_k$ vanishes; therefore only subsets $R \supseteq A^c$ contribute. Write $R = A^c \sqcup U$ with $U \subseteq A$. For $k \in A$ we have $\eta_k = -q^{2k-1}$ and $1 - \eta_k = 1 + q^{2k-1}$, while for $k \in A^c$ the factor $\eta_k = 1$ is harmless. Thus

$$G_\lambda(-q) = \sum_{U \subseteq A} K_{\lambda, \alpha(A^c \cup U)} \prod_{k \in U} (-q^{2k-1}) \prod_{k \in A \setminus U} (1 + q^{2k-1}). \quad (2)$$

Content of $\alpha(A^c \cup U)$. The marks of $\alpha(A^c)$ are $\{0\} \cup A^c \cup \{n\}$. When n is even, $A^c = \{2, 4, \dots, n-2\}$ and the gaps are all equal to 2, giving the multiset (2^m) ; when n is odd, $A^c = \{1, 3, \dots, n-2\}$ and the gaps are $(1, 2^m)$. In either case the content is $(2^m, 1^e)$, and the m parts equal to 2 occupy the gaps whose midpoints are exactly the elements of A . Adjoining an element $k \in A$ to the mark set splits the 2-gap centred at k into two 1's; distinct elements of U split distinct gaps. Hence the content multiset of $\alpha(A^c \cup U)$ is $(2^{m-|U|}, 1^{e+2|U|})$, and

$$K_{\lambda, \alpha(A^c \cup U)} = \left\langle s_\lambda, h_2^{m-|U|} h_1^{e+2|U|} \right\rangle.$$

Substituting into (??) and pulling the pairing outside,

$$G_\lambda(-q) = \left\langle s_\lambda, \sum_{U \subseteq A} h_2^{m-|U|} h_1^{e+2|U|} \prod_{k \in U} (-q^{2k-1}) \prod_{k \in A \setminus U} (1 + q^{2k-1}) \right\rangle.$$

The sum over U factorises over A : for each $k \in A$ the choice $k \in U$ contributes $h_1^2 \cdot (-q^{2k-1})$ and the choice $k \notin U$ contributes $h_2 \cdot (1 + q^{2k-1})$, the common factor h_1^e being independent of U . Therefore

$$G_\lambda(-q) = \left\langle s_\lambda, h_1^e \prod_{k \in A} (h_2(1 + q^{2k-1}) - h_1^2 q^{2k-1}) \right\rangle.$$

Finally $h_2(1 + q^{2k-1}) - h_1^2 q^{2k-1} = h_2 + (h_2 - h_1^2) q^{2k-1} = h_2 - e_2 q^{2k-1}$, using $h_1^2 = h_2 + e_2$. This is the claim. $\square$

Remark 1. At $q = 1$ each factor is $h_2 - e_2 = p_2$, so $G_\lambda(-1) = \langle s_\lambda, h_1^e p_2^m \rangle = \langle s_\lambda, p_1^e p_2^m \rangle = \chi^\lambda(2^m 1^e) = \chi^\lambda(w_0)$, the longest-element character: Proposition ?? contains and re-proves the previously established $q = -1$ fiber.

3 The vanishing law

Two standard facts about the 2-core are used; both are classical (James–Kerber, *The Representation Theory of the Symmetric Group*, §2.7, the theory of the 2-quotient and Littlewood's formula).

Lemma 2 (2-core obstruction). *If $b < |\text{2-core}(\lambda)|$ then $\chi^\lambda(2^a 1^b) = 0$ (where $2a + b = n$).*

Justification. By Murnaghan–Nakayama, $\chi^\lambda(2^a 1^b) = \sum_\mu (\pm) f^\mu$ summed over partitions μ of size b obtained from λ by removing a dominoes. Domino removal preserves the 2-core, so any such μ has the same 2-core as λ and hence $|\mu| = b \geq |\text{2-core}(\lambda)|$. If $b < |\text{2-core}(\lambda)|$ there are no such μ and the sum is empty. $\square$

Lemma 3 (2-quotient value). *Let $c_0 = |2\text{-core}(\lambda)|$ and $w = (n - c_0)/2$. Then*

$$\chi^\lambda(2^w 1^{c_0}) = \varepsilon_\lambda f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{2\text{-core}(\lambda)}, \quad \varepsilon_\lambda \in \{\pm 1\}.$$

In particular this value is nonzero.

Define, for $0 \leq j \leq m$,

$$E_j := \left\langle s_\lambda, (-e_2)^j p_1^e p_2^{m-j} \right\rangle. \quad (3)$$

Lemma 4. $E_j = 0$ for all $j < c(\lambda)$, and $E_c = (-\frac{1}{2})^c \chi^\lambda(2^{m-c} 1^{c_0}) \neq 0$.

Proof. Using $e_2 = \frac{1}{2}(p_1^2 - p_2)$,

$$(-e_2)^j p_1^e p_2^{m-j} = \left(-\frac{1}{2}\right)^j \sum_{a=0}^j \binom{j}{a} (-1)^{j-a} p_1^{2a+e} p_2^{m-a},$$

so that, pairing with s_λ ,

$$E_j = \left(-\frac{1}{2}\right)^j \sum_{a=0}^j \binom{j}{a} (-1)^{j-a} \chi^\lambda(2^{m-a} 1^{2a+e}).$$

The class $2^{m-a} 1^{2a+e}$ has $2a + e$ fixed points. By Lemma ?? its character vanishes whenever $2a + e < c_0 = 2c + e$, i.e. whenever $a < c$. For $j < c$ every index $a \leq j$ satisfies $a < c$, so all terms vanish and $E_j = 0$. For $j = c$ only the top term $a = c$ survives, giving $E_c = (-\frac{1}{2})^c \chi^\lambda(2^{m-c} 1^{c_0})$, which is nonzero by Lemma ??. $\square$

Proof of Theorem ??. Expand Proposition ?? about $q = 1$. Each factor is

$$h_2 - e_2 q^{2k-1} = p_2 - e_2 g_k, \quad g_k := q^{2k-1} - 1 = \sum_{i \geq 1} \binom{2k-1}{i} (q-1)^i,$$

where we used $h_2 - e_2 = p_2$; note g_k has no constant term, $g_k = (2k-1)(q-1) + O((q-1)^2)$. Expanding the product over A ,

$$h_1^e \prod_{k \in A} (p_2 - e_2 g_k) = h_1^e \sum_{S \subseteq A} (-e_2)^{|S|} p_2^{m-|S|} \prod_{k \in S} g_k.$$

Since $\prod_{k \in S} g_k$ is divisible by $(q-1)^{|S|}$, the coefficient of $(q-1)^r$ receives contributions only from $|S| = j \leq r$, and pairing with s_λ ,

$$[(q-1)^r] G_\lambda(-q) = \sum_{j=1}^r E_j \sum_{\substack{S \subseteq A \\ |S|=j}} [(q-1)^r] \prod_{k \in S} g_k, \quad (4)$$

using (??) and $h_1^e = p_1^e$, $\left\langle s_\lambda, (-e_2)^j p_1^e p_2^{m-j} \right\rangle = E_j$. (The $j = 0$ term contributes only to $r = 0$.)

By Lemma ??, $E_j = 0$ for $1 \leq j \leq r$ whenever $r < c$; hence $[(q-1)^r] G_\lambda(-q) = 0$ for all $r < c$. For $r = c$ only $j = c$ survives in (??), and for $|S| = c$ one has $[(q-1)^c] \prod_{k \in S} g_k = \prod_{k \in S} (2k-1)$ (the lowest-order term of each g_k). Summing over S ,

$$[(q-1)^c] G_\lambda(-q) = E_c e_c(\{2k-1 : k \in A\}) \neq 0,$$

by Lemma ?? and positivity of e_c . Since $G_\lambda(q) = \sum_k a_k (q+1)^k$ gives $G_\lambda(-q) = \sum_k a_k (-1)^k (q-1)^k$, we have $[(q-1)^k]G_\lambda(-q) = (-1)^k a_k$. Therefore $a_k = 0$ for $k < c$ and

$$a_c = (-1)^c E_c e_c(\{2k-1 : k \in A\}) = (-1)^c e_c(\{2k-1 : k \in A\}) \left(-\frac{1}{2}\right)^c \chi^\lambda(2^{m-c} 1^{c_0}) \neq 0.$$

Thus $\text{ord}_{q=-1} G_\lambda = c(\lambda)$, and the displayed value is the leading coefficient. Substituting Lemma ?? for $\chi^\lambda(2^{m-c} 1^{c_0})$ and absorbing the sign ε_λ together with the (-1) 's into $\pm$ yields the 2-quotient form, with $S(n, c) = \pm e_c(\{2k-1 : k \in A\}) 2^{-c} f^{2\text{-core}(\lambda)}$ depending only on (n, c) . The normalisations $S(n, 0) = 1$ (then $c = 0$, $e_0 = 1$, core trivial) and $S(n, 1) = -\binom{n}{2}$ (then $e_1(\{2k-1 : k \in A\}) = \binom{n}{2}$) for n odd and $f^{(2,1)} = 2$) are immediate. $\square$

4 The $c = 1$ stratum, worked out

Let λ have 2-core $(2, 1)$, so n is odd, $c_0 = 3$, $c = 1$, $m = (n-1)/2$ and $w = m-1 = (n-3)/2$ dominoes. Here $A = \{2, 4, \dots, n-1\}$ and $e_1(\{2k-1 : k \in A\}) = \sum_{k \in A} (2k-1) = \binom{n}{2}$. Since the 2-core is nonempty, $\chi^\lambda(w_0) = G_\lambda(-1) = 0$ (Lemma ??), so the order is at least 1; Theorem ?? gives order exactly 1 with

$$a_1 = -E_1 \binom{n}{2} = \frac{1}{2} \binom{n}{2} \chi^\lambda(2^{m-1} 1^3) = (-1)^{n(\lambda)+1} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} \binom{n}{2},$$

the last step by Lemma ?? (which here gives $\chi^\lambda(2^{m-1} 1^3) = (-1)^{n(\lambda)+1} \cdot 2 \cdot f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|}$, with $f^{(2,1)} = 2$). For the smallest case $\lambda = (2, 1)$ ($n = 3$, $w = 0$): $G_{(2,1)}(q) = q^3 + 1 = (q+1)(q^2 - q + 1)$, so $a_1 = 3$, matching $(-1)^{1+1} \cdot 1 \cdot 1 \cdot \binom{0}{0} \cdot \binom{3}{2} = 3$.

5 Computational verification

Every claim was checked by independent computation (exact integer / rational arithmetic), with no exceptions, for all partitions in the indicated range.

- **Master identity (Proposition ??).** The polynomial identity $G_\lambda(-q) = \langle s_\lambda, h_1^e \prod_{k \in A} (h_2 - e_2 q^{2k-1}) \rangle$, with the right side expanded into power sums and evaluated by Murnaghan–Nakayama, agrees with direct SYT enumeration of G_λ for all $\lambda \vdash n$, $2 \leq n \leq 9$.
- **Order law and leading coefficient.** For all $\lambda \vdash n$, $4 \leq n \leq 12$ (265 shapes), $\text{ord}_{q=-1} G_\lambda = c(\lambda)$, and the leading Taylor coefficient a_c equals $(-1)^c e_c(\{2k-1 : k \in A\}) (-\frac{1}{2})^c \chi^\lambda(2^{m-c} 1^{c_0})$.
- **E_j vanishing (Lemma ??) and the 2-core obstruction (Lemma ??).** $\chi^\lambda(2^a 1^b) = 0$ for all $b < |2\text{-core}(\lambda)|$, and $E_j = 0$ for $j < c$, for all $\lambda \vdash n \leq 12$.
- **$S(n, c)$.** The reduced leading coefficient $a_c / ((-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|})$ is single-valued for each (n, c) with $n \leq 12$, and equals $\pm e_c(\{2k-1 : k \in A\}) 2^{-c} f^{2\text{-core}}$ (e.g. $S(n, 1) = -\binom{n}{2}$, $S(6, 3) = 90$, $S(8, 3) = 1624$, $S(10, 5) = 238680$, $S(12, 5) = 7490736$).

Scripts. ~/projects/scratch/prove-2026-06-01-graded-2core/ (notably `verify_master.py`, `verify_general.py`, `verify_full.py`, `referee_check.py`).